

The Menstrual Cycle

Learning objective

At end of this session student be able to

- Define menstrual cycle
- Classify Ovarian cycle
- Explain Female Hormonal System

Function of Female Reproductive System

- Produce sex hormones
- Produce functioning gametes [ova]
- Support & protect developing embryo



- + **Puberty**: the period when the endocrine and gametogenic functions of the gonads first develop to the point where reproduction is possible.
- + **Thelarche**-the development of breasts, is the first event.
- + **pubarche** -the development of axillary and pubic hair.
- + **Menarche**-the first menstrual period
Age is variable (8-13years)

Hypothalamo-pituitary-ovarian (HPO) axis at different stage of life

- **Fetal life**- active and functional from 20 weeks of life.
- **Infancy and childhood**- high level of FSH and LH at birth gradually decline and minimum level achieved by two years age.
- **Pre-puberty**- hypothalamus is very much sensitive to negative feedback by even by small amount of oestrogen.
- FSH and LH secretions are inhibited.
- **Puberty**- hypothalamus becomes more insensitive to oestrogen negative feedback. Hence, increased amount of GnRH, FSH and LH are secreted, which intern stimulates the ovary to secrete oestrogen and progesterone. This eventually results in telarche, adrenarche, and menarche

Cont...

- During all the reproductive years of adult life, between about **13 and 46 years of age**, 400 to 500 of the primordial follicles develop enough to expel their ova—one each month;
- The remainder degenerate (become *atretic*).

Physiology of the Female Reproductive Organs

Puberty - the age of sexual maturation

- This is the stage of life at which secondary sexual characteristics appear.
- Maturation of reproductive organs occur at approximately at the of age 12 to 13 years
- In girls pubertal changes typically occur in the order of:
 - Marked physical growth
 - Increase in the transverse diameter of the pelvis
 - Breast development
 - Growth of pubic and axillary hair
 - Vaginal secretion /Menarche

Female Hormonal System

✓ The female hormonal system consists of three hierarchies of hormones, as follows:

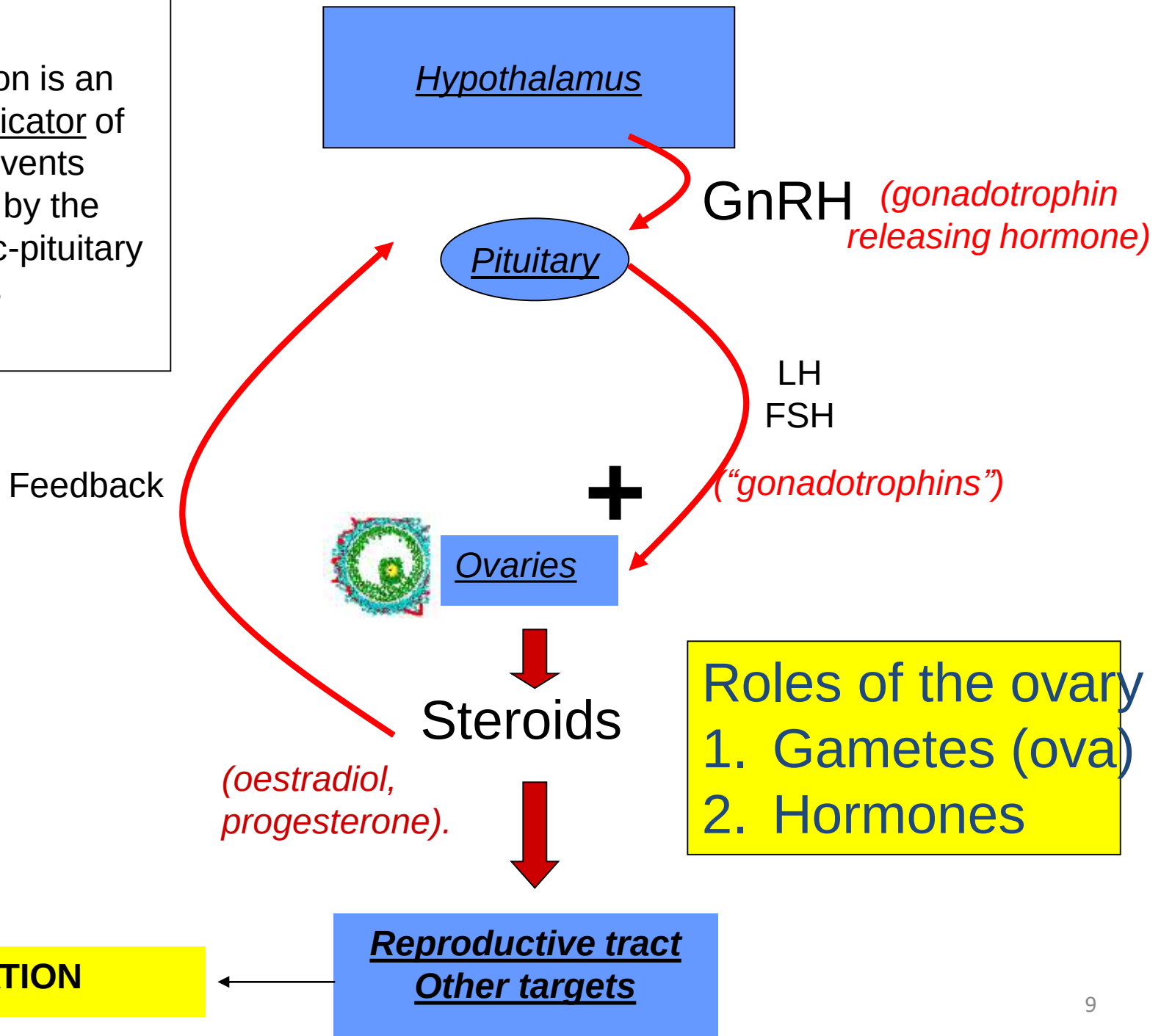
1. A hypothalamic releasing hormone, *gonadotropin-releasing hormone*

(*GnRH*)

2. The anterior pituitary sex hormones, *follicle-stimulating hormone (FSH)* and *luteinizing hormone (LH)*, both of which are secreted in response to the release of GnRH from the hypothalamus

3. The ovarian hormones, *estrogen and progesterone*, which are secreted by the ovaries in response to the two female sex hormones from the anterior pituitary gland

Menstruation is an external indicator of ovarian events controlled by the hypothalamic-pituitary axis



Menstruation

- This is a monthly (in average) flow of blood through the vagina as a result of endometrial shedding.
- The length of normal menstrual cycle is determined by counting from the 1st day of period to the 1st day of menses the following month.
- Normally menstrual period varies 21-- 35days, but the average interval is 28 days
- Duration of menstruation is variable, but 2-8 days is normal
- Blood loss varies from spotting to heavy bleeding, but normal blood loss is in b/n 30ml - 80ml - more than 80ml is considered as excessive & less than 30ml considered spotting bleedings.

Menstruation cont...

- The average age at which menarche (the first menstrual period) occurs at the average age of 12.8 years.
- This may occur as early as age 9 or as late as age 17 years.
- Four body structures that are involved in the normal physiology of the menstrual cycle are:
 - ✓ The hypothalamus
 - ✓ The pituitary gland
 - ✓ The ovaries and
 - ✓ The uterus.

Menstrual phase -

- Is characterized by vaginal bleeding
- Lasts 3-5 days
- If the ovum is not fertilized the anterior pituitary gland withdraw the LH after 14-15 days
- Preparation for pregnancy ceases and the endometrium sheds with the menstrual flow.

Ovarian Cycle

- Monthly series of events associated with the maturation of an egg
- **Follicular phase** – period of follicle growth (days 1–14)
- **Luteal phase** – period of corpus luteum activity (days 14–28)
- Ovulation occurs mid-cycle

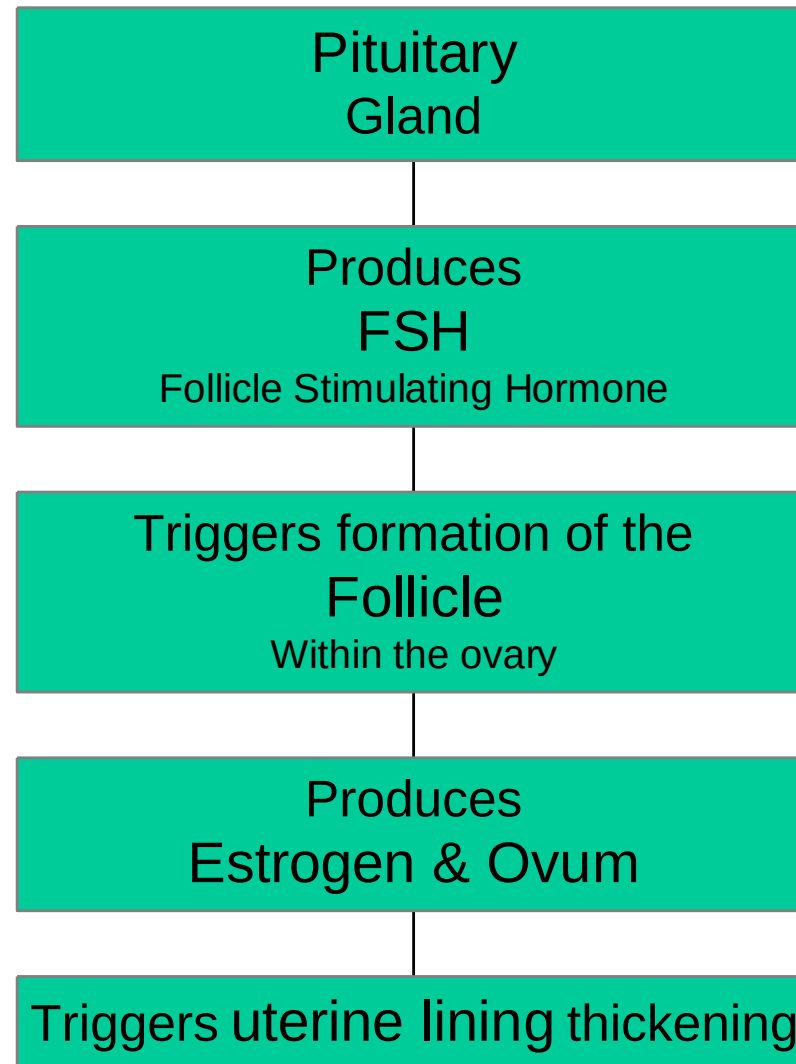
Follicular Phase (*Proliferative phases*)

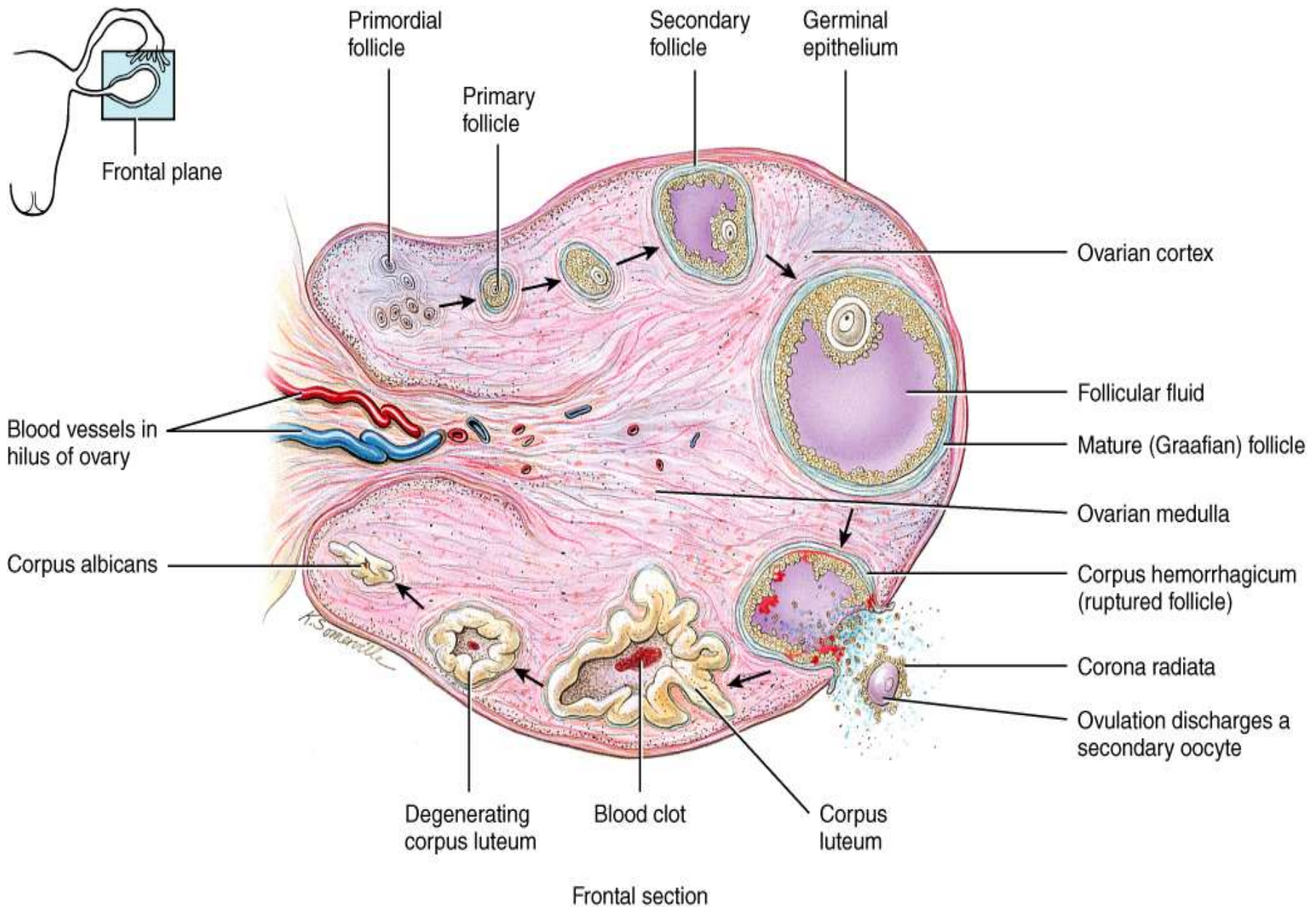
- Follows menstruation
- Is under the control of estrogen
- Consists of re-growth & thickening of endometrium.
- *The ovum surrounded by a single layer of **granulosa cells** is called a **primordial follicle***
- The primordial follicle, directed by the oocyte, becomes a primary follicle
- The anterior pituitary releases the follicle stimulating hormone (FSH).
- Causes the graffian follicle to increase in size this leads to rise in the estrogen level.
- Finally this causes new growth of the endometrium

Follicular Phase

- **Primary follicle becomes a secondary follicle**
 - The theca folliculi and granulosa cells cooperate to produce estrogens
 - The zona pellucida forms around the oocyte
 - The antrum is formed
- **The secondary follicle becomes a vesicular follicle**
 - The antrum expands and isolates the oocyte and the corona radiata
 - The full size follicle (vesicular follicle) bulges from the external surface of the ovary
 - The primary oocyte completes meiosis I, and the stage is set for ovulation

Days 1-14





Ovarian Cycle:

follicular phase

FSH and LH in the Follicular phase

LH surge

ovulation

LH surge lasts 48 h

Inc GnRH bursts

FSH
LH

1

4

14

28

proliferative phase
(11 d)

secretory phase (12d)

Endometrial Cycle:

menstrual

Ovulation

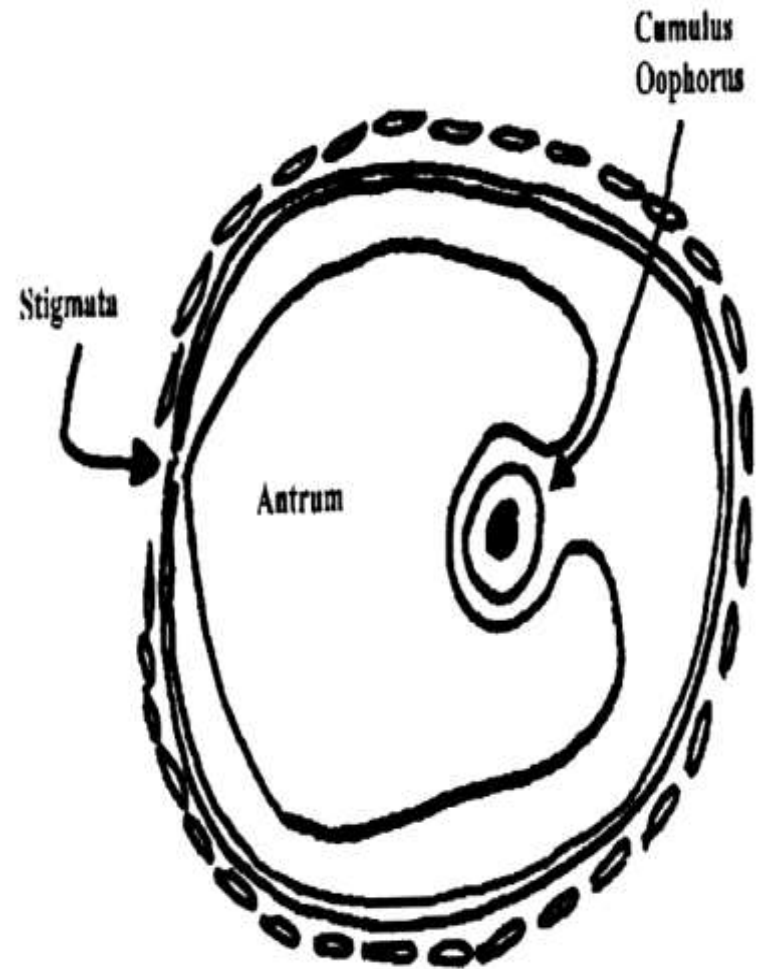
- Ovulation occurs when the ovary wall ruptures and expels the secondary oocyte (14 days)
- it is release of egg from graffian follicle.
- During this time the FSH, LH & estrogen found the peak level.
- When the graffian follicle raptures the egg will release & the estrogen level drops.
- ↑ Estrogen → LH "surge" & FSH spike → egg release
- Inhibit↑ pushes FSH down , ↓ new follicle development

LH surge

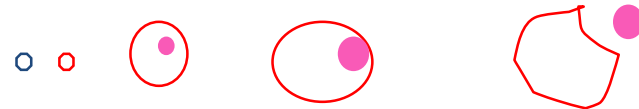
- Lasts for 48 hours
- Ovulation occurs after **36 hours**
- Accompanied by rapid fall in estradiol level
- Affects follicular wall → follicular rupture
- Granulosa cell → lutenization → progesterone synthesis

Ovulation

- The stigmata, a cone-shaped protrusion of the follicle wall, appears just before ovulation
- Extensive follicular vascularization occurs immediately prior to ovulation
- Gentle release of the oocyte



Ovarian Cycle:



follicular phase

ovulation

LH → surge

Increase in estradiol to stimulate LH surge. Then estradiol has Negative feedback on GnRH to reduce LH, FSH.

estradiol

FSH

LH

neg feedback--GnRH

1

4

14

28

Endometrial

Proliferative phase (11 d)

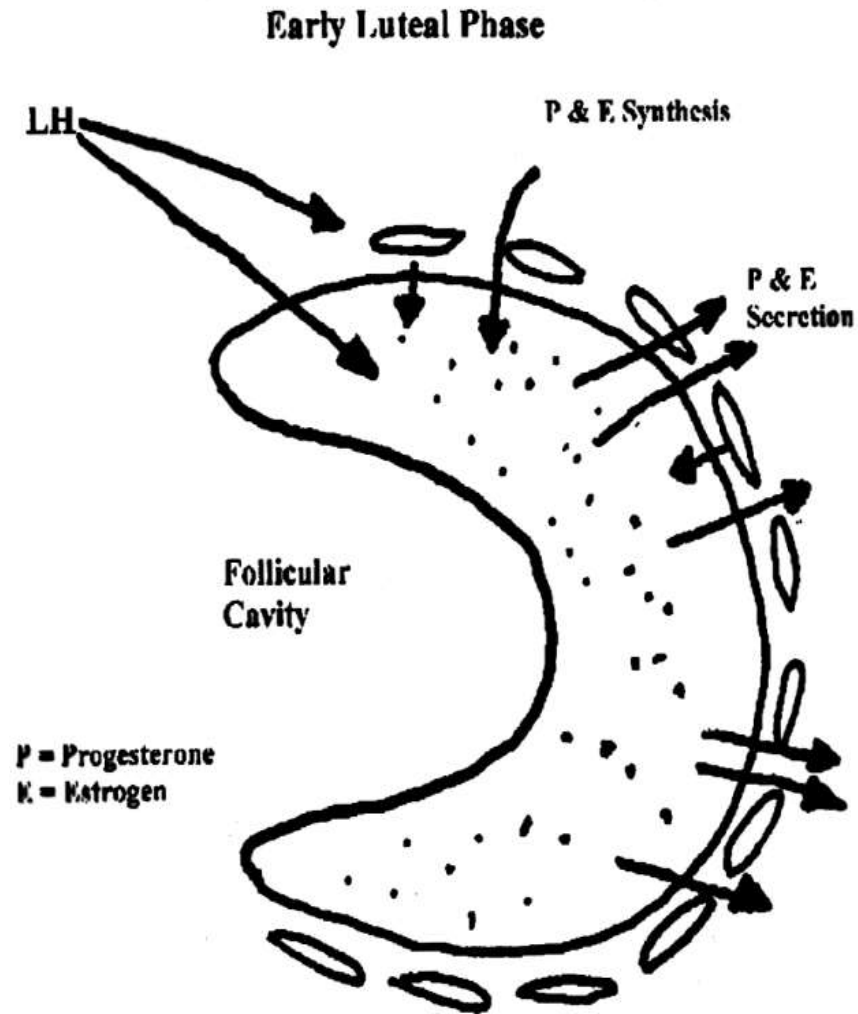
Secretory phase (12d)

Cycle: menstrual

9/5/2021

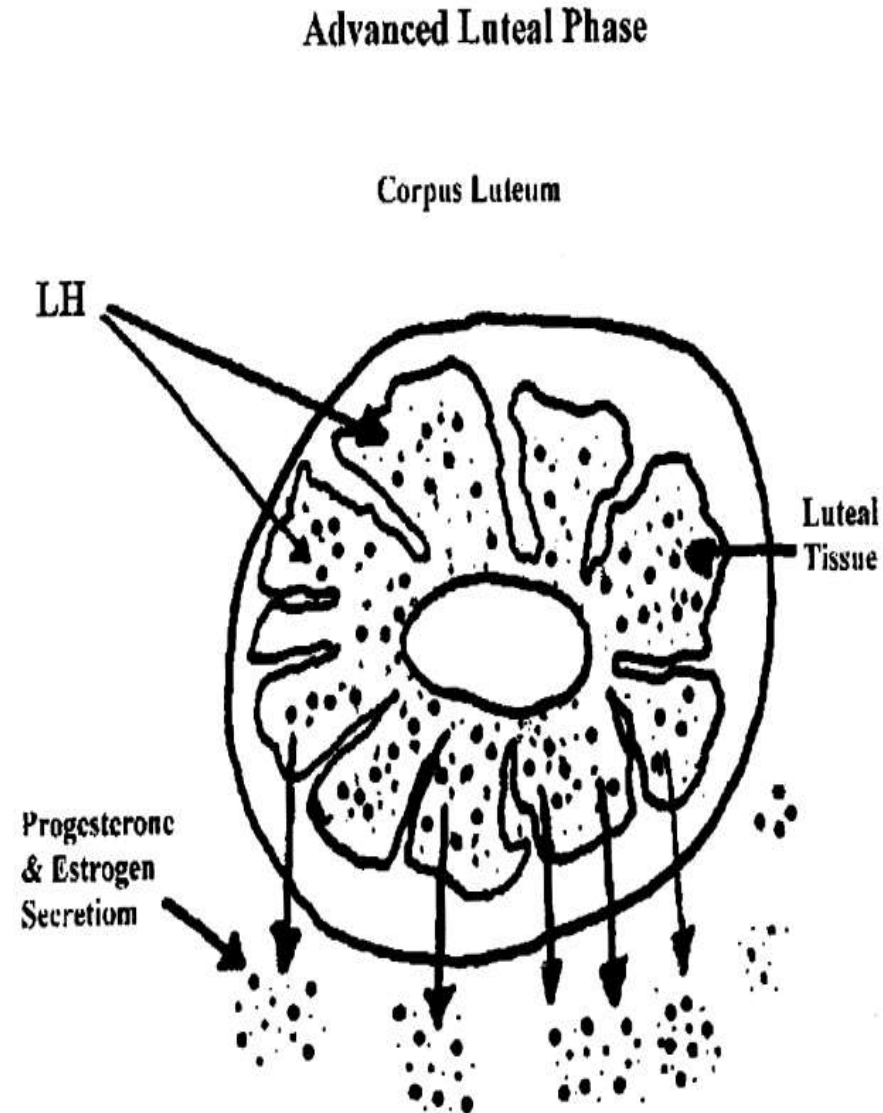
Corpus Luteum Development 1

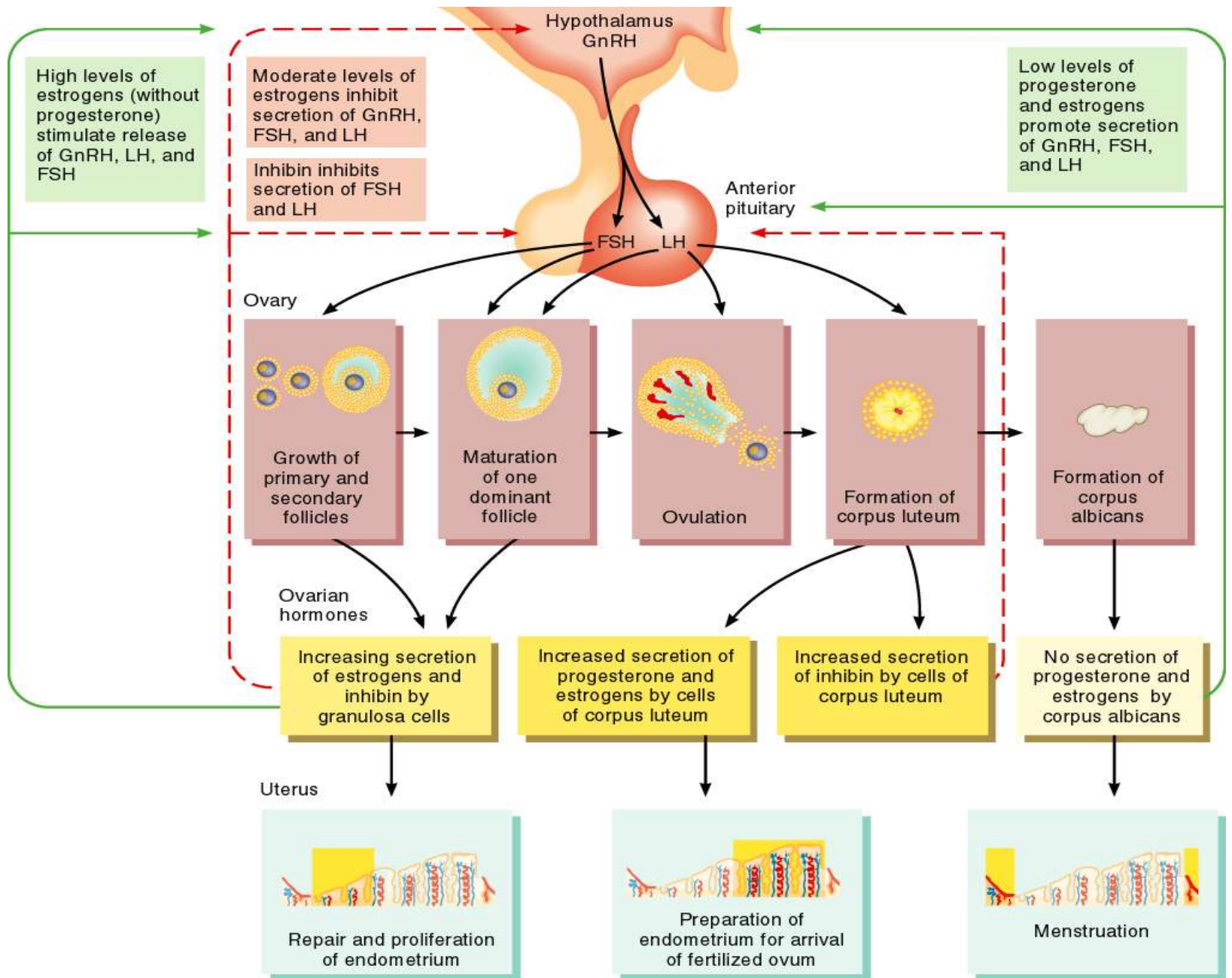
- LH stimulates thecal & granulosa cell invasion of follicular cavity
- LH induces the luteinization of both cell types.
- Progesterone synthesis increases as number of luteal cells increase



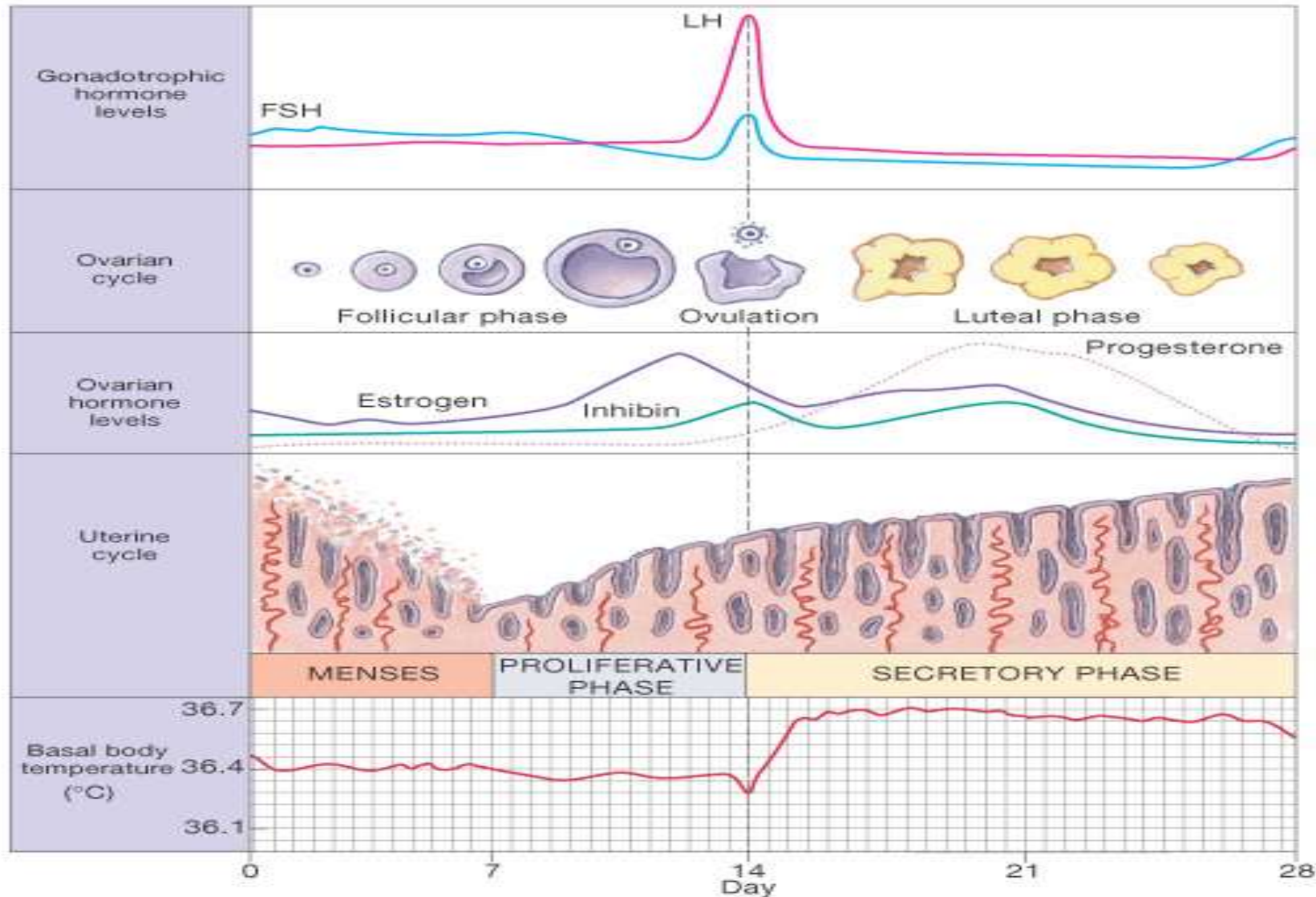
Corpus Luteum Development 2

- LH stimulates increased **P & E** secretion
- Progesterone actions
 - Suppress follicular maturation in the ipsilateral ovary
 - Thermo-genic activity
 - Endometrial maturation





Overview of the Menstrual Cycle



Luteal phase

- ✓ The luteal (secretory, progesterone) phase, lasting approximately 13 days, coincides with the formation, functioning, and growth of the corpus luteum.

The endometrium becomes thicker (5-7mm.) and forms
3layers:

- ✓ Superficial compact layer
- ✓ Middle spongy layer
- ✓ Deep basal layer.
- ✓ The cells become **columnar**.
- ✓ The uterine glands become more **convoluted and full of** secretion (mucous and glycogen)

Cont...

- The spiral arteries (supplying the compact and spongy layers) become dilated and tortuous.
- Direct **arterio venous anastomosis** are **prominent features of** this stage

N.B This stage is under the effect of the hormone **progesterone** secreted by **the corpus luteum**.

➤ **If fertilization does not occur:**

1. The corpus luteum degenerates
2. Estrogen and progesterone levels fall and the secretory endometrium enters an **ischemic phase**
3. Menstruation occurs.

Ischemic Phase

- The ischemic phase occurs when the **oocyte** is not fertilized.
- Ischemia (reduced blood supply) occurs as the spiral arteries constrict, giving the **endometrium a pale** appearance.
- This constriction results from the decreasing secretion of hormones, primarily progesterone, by the **degenerating corpus luteum**.
- In addition to **vascular changes**, the hormone **withdrawal** results in the **stoppage of glandular secretion, a loss of interstitial fluid, and a marked shrinking of the** endometrium.

Cont....

- Toward the end of the ischemic phase, the spiral arteries become **constricted for longer periods**.
- This results in ***venous stasis and patchy ischemic necrosis*** (death) in the superficial tissues. Eventually, rupture of damaged vessel walls follows and blood seeps into the surrounding connective tissue.
- Small pools of blood form and break through the endometrial surface, resulting in bleeding into the uterine lumen and from the vagina.

Cont...

- As small pieces of the endometrium detach and pass into the uterine cavity, the torn ends of the arteries bleed into the uterine cavity, resulting in a loss of **20 to 80 mL of blood**.
- Eventually, over **3 to 5 days, the entire compact layer and most of the spongy layer of the endometrium are** discarded in the *menses*.
- Remnants of the spongy and basal layers remain to undergo regeneration during the subsequent **proliferative phase of the endometrium**.

If fertilization occurs

- Cleavage of the zygote and blastogenesis (formation of blastocyst) occur.
- The blastocyst begins to implant in the endometrium on approximately the **sixth day of the luteal phase (day 20 of a 28-day cycle)**.
- Human chorionic gonadotropin, a hormone produced by the syncytiotrophoblast, keeps the corpus luteum secreting estrogens and progesterone.
- The luteal phase continues and menstruation does not occur.